UNIVERSITAT POLITÈCNICA DE CATALUNYA
BARCELONATECH
Departament d'Enginyeria Telemàtica

AI-based Intrusion Detection for IoMT environments

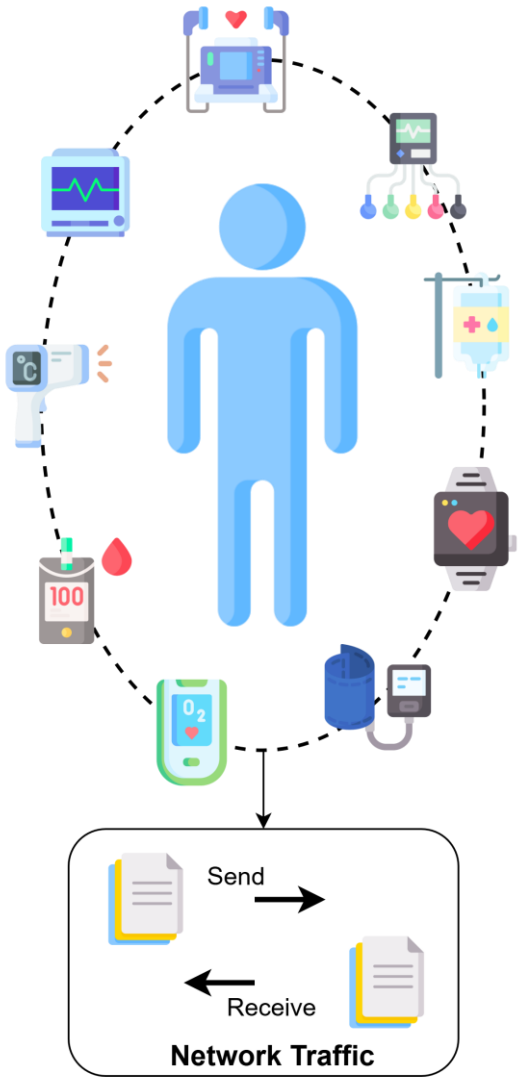
Thesis Offer to Pantelis Petrakis

Advisors: Prof. Dr. Josep Pegueroles and PhD Candidate Jordi Domènech

28/11/2025

Motivation

Internet of Medical Things (IoMT)



Personal data is being sent

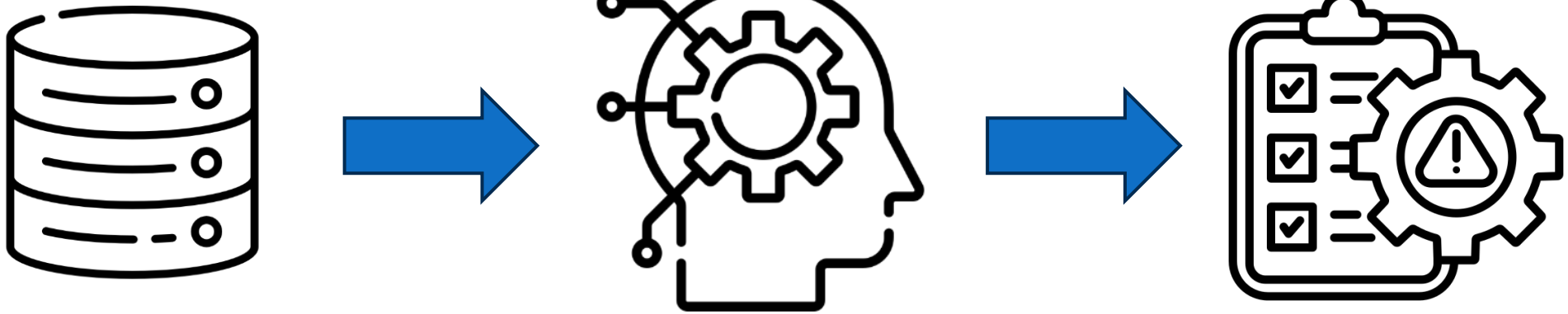
Devices are connected to patients

POTENTIAL INTEREST FOR CYBERCRIMINALS

Cyber attacks increased by... **55%** [1]

[1] E. U. A. for Cybersecurity, M. Theocharidou, and I. Lella, ENISA threat landscape report : health sector (January 2021 to March 2023). European Network and Information Security Agency, 2023.

Objective



DATA COLLECTION & PREPROCESSING

AI-BASED IDS

EVALUATION & ANALYSIS



CICIoMT2024, IoMT-TrafficData, and our own IoMT dataset

[2] J. Doménech, I. V. Martin-Faus, S. Mhiri, and J. Pegueroles, "Ensuring patient safety in IoMT: A systematic literature review of behavior-based intrusion detection systems," Internet of Things, vol. 28, p. 101420, 2024.

[3] J. Doménech, O. León, M. S. Siddiqui, and J. Pegueroles, "Evaluating and enhancing intrusion detection systems in IoMT: The importance of domain-specific datasets," Internet of Things, vol. 32, p. 101631, 2025.

[4] J. Doménech, S. Mhiri, M. Shuaib Siddiqui and J. Pegueroles, "Preventive and Reactive Cybersecurity Techniques on IoT Devices in Healthcare Environments," 2025 IEEE 11th International Conference on Network Softwarization (NetSoft), Budapest, Hungary, 2025, pp. 261-264, doi: 10.1109/NetSoft64993.2025.11080634.

[5] O. León, I. V. Martin-Faus, J. Pegueroles, and J. Doménech, "Herramienta de preprocesamiento de tráfico iomt para generación configurable de datasets, in Actas de las XVII Jornadas de Ingeniería Telemática (JITEL 2025), ser. JITEL 2025. Cáceres (España): Universidad de Extremadura, 2025.

[6] Alarcón, A., León, O., Pegueroles, J., & Doménech, J. (2025). Técnicas de Machine Learning para la detección de ciberataques en redes IoT médicas. Retrieved from https://hdl.handle.net/10612/24562.

Problem Statement

Most existing IDS models for IoMT are trained on **limited or non-representative datasets**, leading to issues such as **overfitting**, **low detection rates** for unknown attacks, and **poor adaptability** to real-world scenarios.

The **lack of a comprehensive approach** that combines **multiple data sources** reduces the system's performance and its practical applicability.

Research Question



How can the **integration of multiple heterogeneous datasets** improve the **performance** and **generalizability** of an ML model in detecting intrusions in IoMT environments?



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AND NOW...
YOUR TURN!



